Properties of OLS and omitted variables: worksheet

$\hat{\beta}_1 = \dfrac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2}$ $\quad t = \dfrac{\hat{\beta}_1 - \beta_{1,0}}{\mathrm{SE}(\hat{\beta}_1)}$ $\quad$ 95% CI: $\hat{\beta}_1 \pm 1.96\ \mathrm{SE}(\hat{\beta}_1)$ $\quad \mathrm{SE}(\hat{\beta}_1) \approx \dfrac{SER}{\sqrt{n} \times s_X}$

Omitted W: $\hat{\beta}_1 \xrightarrow{p} \beta_1 + \beta_2 \cdot \dfrac{\mathrm{cov}(X_i, W_i)}{\mathrm{var}(X_i)}$, where β_2 is W's effect on Y holding X fixed.

Problem 1. (From the mean to the slope) Fill in the column for the regression slope.

	Learning about a mean	**Learning about a slope**
Population parameter	μ_Y	
Estimator	$\bar{Y} = \frac{1}{n}\sum_i Y_i$	
Unbiased? When?	$E[\bar{Y}] = \mu_Y$, in a random sample	
Consistent? When?	$\bar{Y} \xrightarrow{p} \mu_Y$, in a random sample	
In large samples	$\bar{Y} \approx N(\mu_Y,\ \sigma_Y^2/n)$	
Standard error	$\mathrm{SE}(\bar{Y}) = s_Y/\sqrt{n}$	
t-statistic	$t = (\bar{Y} - \mu_0)/\mathrm{SE}(\bar{Y})$	
p-value	$2\Phi(-\lvert t\rvert)$	
95% confidence interval	$\bar{Y} \pm 1.96\ \mathrm{SE}(\bar{Y})$	

Problem 2. (True or false? Explain in one sentence)

(a) If $E[u_i \mid X_i] = 0$, the $\hat{\beta}_1$ from our sample equals β_1.

(b) If $\mathrm{cov}(X_i, u_i) \neq 0$, a sample of 10 million observations gives a $\hat{\beta}_1$ very close to β_1.

(c) $\mathrm{SE}(\hat{\beta}_1)$ tells us how far $\hat{\beta}_1$ is likely to be from β_1 because of omitted variables.

(d) A 90% confidence interval for β_1 is narrower than a 95% interval built from the same data.

(e) β_1 must be inside our 95% confidence interval.

(f) Two different random samples give the same 95% confidence interval for β_1.

Problem 3. (What makes the standard error small?) Use $\mathrm{SE}(\hat{\beta}_1) \approx SER/(\sqrt{n} \times s_X)$.

(a) What happens to $\mathrm{SE}(\hat{\beta}_1)$ if the sample is four times as large?

(b) What happens if the spread of X, s_X, doubles? Explain in words why a wider range of X helps.

(c) What happens if the typical miss around the line, the SER, halves?

(d) You want to estimate how class size affects test scores, and can afford one of two samples: (A) 400 schools whose average class sizes have a standard deviation of 2 students, or (B) 100 schools whose class sizes have a standard deviation of 6 students. Assuming the same SER, which sample gives the smaller standard error, and by how much?

Problem 4. (Do older workers earn more?) We take a random sample of 5,000 formal jobs held by workers aged 35–64 in Brazil. `pay` is December pay in reais per month, `age` is the worker's age in years, and `educ` is years of schooling.

```
. regress pay age, robust

Linear regression                               Number of obs     =      5,000
                                                F(1, 4998)        =       0.88
                                                Prob > F          =     0.3479
                                                R-squared         =     0.0002
                                                Root MSE          =     4136.7

------------------------------------------------------------------------------
             |               Robust
         pay | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         age |  -7.795113     8.3039    -0.94   0.348     -24.0744    8.484174
       _cons |    4395.14   373.0014    11.78   0.000     3663.893    5126.386
```

(a) Interpret the slope. How much does pay change over ten years of age?

(b) Test H_0: $\beta_1 = 0$ at the 5% level.

(c) Give the 95% confidence interval for the change in pay over ten years of age.

(d) Older Brazilians went to school when schooling was less common. Suppose `educ` is the omitted variable W. Sign β_2 and $\operatorname{cov}(X, W)$, and say which way $\hat{\beta}_1$ is biased.

Now we add `educ`, and we also regress `educ` on `age`:

```
. regress pay age educ, robust

Linear regression                               Number of obs     =      5,000
                                                F(2, 4997)        =     243.12
                                                Prob > F          =     0.0000
                                                R-squared         =     0.1394
                                                Root MSE          =     3838.3

------------------------------------------------------------------------------
             |               Robust
         pay | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         age |   27.19078   8.249121     3.30   0.001     11.01888    43.36267
        educ |   483.1861   22.66198    21.32   0.000     438.7587    527.6136
       _cons |  -2886.414   512.9361    -5.63   0.000    -3891.994   -1880.835
```

```
. regress educ age, robust

Linear regression                               Number of obs     =      5,000
                                                F(1, 4998)        =     122.39
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0275
                                                Root MSE          =     3.1948

------------------------------------------------------------------------------
             |               Robust
        educ | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         age |  -.0724066   .0065448   -11.06   0.000    -.0852374   -.0595759
       _cons |   15.06987   .2912711    51.74   0.000     14.49886    15.64089
```

(e) Did the coefficient on `age` move as you predicted in (d)? Interpret it, and test whether it is zero.

(f) Use the omitted variable bias formula and the last output to recover the short-regression slope from the long one.

Problem 5. (More data does not fix it) A researcher regresses students' GPA on the hours per week they work at a part-time job. Suppose that, in truth, each extra hour of work lowers GPA by $\beta_1 = -0.01$. Family income, W in $10,000s, also matters: holding hours fixed, each $10,000 raises GPA by $\beta_2 = 0.02$. Students from poorer families work more: $\mathrm{cov}(X_i, W_i)/\mathrm{var}(X_i) = -0.5$, so each extra hour of work goes with $5,000 less family income. The researcher leaves family income out.

(a) Where does $\hat{\beta}_1$ land in large samples? Does the regression overstate or understate the harm of working?

(b) The researcher has 100 students; a colleague has 100,000. How do the two estimates differ: in where they are centered, in how spread out they are, or both?

(c) The colleague reports $\hat{\beta}_1 = -0.020$ with a standard error of 0.001 and concludes: "Each hour of work lowers GPA by 0.02, and our estimate is very precise." What is right and what is wrong with this claim?